

Contents

HAR-RS-DOW Variance Forecasting for Bitcoin: Statistical Dominance and Applications in Risk

Management and Option Pricing	1
Abstract	1
1. Introduction	2
2. Literature	3
3. Methodology	3
4. Data and realised-variance estimator	5
5. In-sample estimation results	5
6. Out-of-sample forecast evaluation	6
7. Application: Value-at-Risk and Expected Shortfall	7
8. Application: Black-Scholes option pricing	8
9. Robustness	9
10. Conclusion	9
References	10

HAR-RS-DOW Variance Forecasting for Bitcoin: Statistical Dominance and Applications in Risk Management and Option Pricing

Sake Saakstra *MSc Econometrics and Operations Research, Financial Track Vrije Universiteit Amsterdam*

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Abstract

This paper studies the daily realized variance dynamics of Bitcoin (BTC-EUR) over the period April 2019 to May 2026 and establishes the Heterogeneous Autoregressive model with Realized Semivariance and Day-of-Week effects (HAR-RS-DOW) as the statistically dominant variance forecaster for this market. Using realized kernel estimation at five-minute frequency, we estimate ten HAR-family specifications on 2 312 in-sample days and evaluate them out-of-sample over 1 312 days. HAR-RS-DOW dominates the next-best alternative by $\Delta\text{BIC} = 490$ in-sample and by 14% lower CRPS out-of-sample, with Diebold-Mariano statistics rejecting equal predictive accuracy against all four nearest competitors at $p < 0.005$. The dominance is robust across forecast horizons ($h \in \{1, 5\}$) and across sub-periods. We further demonstrate two canonical applications. First, HAR-driven density forecasts produce conservatively calibrated Value-at-Risk and Expected Shortfall under three density specifications (Normal, Student-t, Hansen 1994 skewed-t), with Acerbi-Szekely Z1 statistics in $[+1.73, +2.11]$ —adequate for Basel III regulatory capital reservation. Second, using HAR- σ as the volatility input to Black-Scholes for at-the-money straddles yields fair-value pricing closest to market-clearing among three vol-input specifications. These results support the literature view that HAR-family models are optimally deployed as variance forecasters underpinning risk and derivative-pricing applications, rather than as direct components of directional trading systems.

JEL classification: C22, C53, C58, G17

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1. Introduction

The forecasting of asset return variance is among the most consequential problems in empirical finance. Variance forecasts are direct inputs to value-at-risk (VaR) and Expected Shortfall (ES) calculations mandated under the Basel III Fundamental Review of the Trading Book (Basel Committee, 2019), to option pricing through Black-Scholes (1973) and its extensions, and to risk-budgeting in portfolio construction. Among the broad family of volatility forecasting models—stochastic volatility, GARCH-type, and realized-measure-based—the Heterogeneous Autoregressive (HAR) model introduced by Corsi (2009) has achieved a singular position: it combines a transparent linear structure with the empirical ability to approximate the long-memory properties of realized volatility (RV), without resorting to the parametric complexity of fractionally integrated specifications.

Bitcoin presents an attractive but underexplored test bed for HAR-type modelling. The market is continuous (24/7), high-volume, has substantial high-frequency data availability, and exhibits both classical features of risky-asset returns (volatility clustering, fat tails, leverage effects) and idiosyncratic features (pronounced day-of-week seasonality, sensitivity to retail-investor flows). Existing HAR applications to crypto markets (Bouri et al., 2017; Yu, 2019; Shen, Urquhart, & Wang, 2020) have established that HAR variants outperform GARCH-type benchmarks, but have not systematically compared the full HAR family with semivariance decomposition (Patton & Sheppard, 2015), quarticity correction (Bollerslev, Patton, & Quaedvlieg, 2016), and day-of-week dummies on a single dataset, nor evaluated the implications for downstream risk and derivative-pricing applications under harmonised density assumptions.

This paper makes three contributions:

1. **A statistically dominant variance forecaster for BTC-EUR.** We compare ten HAR-family specifications on a 2 312-day in-sample window and a 1 312-day out-of-sample window. The HAR-RS-DOW specification—Heterogeneous Autoregressive with positive and negative Realized Semivariance components and six day-of-week dummies—dominates all alternatives across in-sample BIC, out-of-sample CRPS, QLIKE, RMSE, and R^2_{OOS} , with Diebold-Mariano (1995) test rejection of equal predictive accuracy at $p < 0.005$ against all four nearest competitors. The dominance is robust across forecast horizons ($h \in \{1, 5\}$ days) and across the early (2022-10 to 2024-01) and late (2024-01 to 2026-05) sub-samples.
2. **Conservatively calibrated Basel III risk forecasts.** We use the HAR-RS-DOW conditional density (Hansen 1994 skewed-t residuals) to construct one-day-ahead Value-at-Risk and Expected Shortfall forecasts at $\alpha \in \{0.01, 0.05\}$. We backtest Expected Shortfall using the Acerbi and Szekely (2014) Z1 statistic across three density specifications: Normal, standardised Student-t with $\nu = 3$, and Hansen 1994 skewed-t with empirically estimated $\nu = 4.4$, $\lambda = +0.01$. The Z1 statistic is positive (conservative) under all three specifications and at both significance levels, with values ranging from +1.73 to +2.11. This implies the HAR-RS-DOW model produces variance forecasts adequate to support regulatory capital reservation under the Basel III FRTB framework.
3. **Fair-value option pricing.** We use HAR-RS-DOW’s conditional volatility forecasts as the input to a Black-Scholes valuation of seven-day at-the-money BTC straddles. Across 1 305 forward-looking valuations, HAR-driven pricing produces an average premium of €10.50 against a realised payoff of €10.30 (edge = €0.20), substantially closer to market-clearing than constant-volatility pricing (edge = €0.86) or rolling 30-day realised-volatility pricing (edge = €0.29). The result is operationally relevant for derivative market-makers and demonstrates HAR-RS-DOW’s value in the canonical option-pricing domain.

The remainder of the paper is organised as follows. Section 2 reviews the HAR family literature and positions

our contribution. Section 3 introduces the HAR-RS-DOW model and the conditional density specification. Section 4 describes the data and the realised-variance estimator. Section 5 presents in-sample estimation results. Section 6 reports the out-of-sample forecast evaluation. Section 7 develops the Value-at-Risk and Expected Shortfall applications. Section 8 presents the option-pricing application. Section 9 reports robustness checks. Section 10 concludes.

2. Literature

The HAR model originates with Corsi (2009), who proposed a parsimonious approximation to the long-memory dynamics of realised volatility through three temporal aggregates—daily, weekly, and monthly—linked to the Heterogeneous Market Hypothesis (Müller et al., 1997). Corsi’s specification proved a highly effective forecasting tool while remaining linear in parameters and amenable to ordinary-least-squares estimation. Subsequent contributions extended the framework along several axes.

Patton and Sheppard (2015) decomposed realised variance into positive and negative semivariance components (RS^+ , RS^-) and demonstrated that the negative-semivariance loading is systematically larger than the positive in equity returns—a leverage effect operating at the variance level. Bollerslev, Patton, and Quaedvlieg (2016) showed that the HAR-Q specification, which weights past observations by realised quarticity to correct for measurement error, improves forecasting accuracy across a wide range of assets. The inclusion of exogenous covariates and calendar effects (day-of-week, macro announcements) is a standard practice in applied volatility forecasting (Andersen, Bollerslev, & Diebold, 2007).

For cryptocurrency markets, the HAR family has been deployed by Bouri et al. (2017) for daily Bitcoin volatility forecasting against GARCH-type alternatives, by Yu (2019) for long-memory characterisation, and by Shen, Urquhart, and Wang (2020) for asymmetric volatility modelling. These studies establish that HAR-type models outperform their GARCH counterparts in crypto contexts. However, no prior study has, to our knowledge, evaluated the full ten-model HAR family in a single unified framework on Bitcoin and traced the implications for risk-management and derivative-pricing applications under harmonised density assumptions.

The choice of conditional density for variance-forecast residuals matters substantially for risk applications. The skewed-t density of Hansen (1994) accommodates both heavy tails and asymmetry while remaining tractable; it has become a workhorse density in financial econometrics (Bauwens, Laurent, & Rombouts, 2006). For Expected Shortfall backtesting, we rely on the Z1 statistic of Acerbi and Szekely (2014), which has emerged as a standard tool for evaluating ES forecasts under Basel III FRTB.

3. Methodology

3.1 The HAR-RS-DOW specification

Let RV_t denote the realised variance on day t , computed from intraday returns at frequency Δ . Denote by RS_t^+ and RS_t^- the positive and negative semivariance components, defined as

$$RS_t^+ = \sum_{i:r_{t,i}>0} r_{t,i}^2, \quad RS_t^- = \sum_{i:r_{t,i}<0} r_{t,i}^2, \quad RV_t = RS_t^+ + RS_t^-.$$

Let $D_{k,t}$ for $k \in \{1, 2, \dots, 6\}$ denote day-of-week dummies indicating Monday through Saturday (Sunday is the omitted baseline). The HAR-RS-DOW model is specified as

$$\log RV_{t+1} = \beta_0 + \beta_d^+ \log RS_t^+ + \beta_d^- \log RS_t^- + \beta_w \overline{\log RV}_{t,t-4} + \beta_m \overline{\log RV}_{t,t-21} + \sum_{k=1}^6 \gamma_k D_{k,t+1} + \varepsilon_{t+1},$$

with weekly and monthly aggregates

$$\overline{\log RV}_{t,t-4} = \frac{1}{5} \sum_{i=0}^4 \log RV_{t-i}, \quad \overline{\log RV}_{t,t-21} = \frac{1}{22} \sum_{i=0}^{21} \log RV_{t-i}.$$

Working on the logarithm of RV ensures approximate Gaussianity of the dependent variable (verified empirically in Section 4) and yields a multiplicative specification in the level. Parameter estimation is by ordinary least squares with Newey-West (1987) heteroskedasticity-and-autocorrelation-consistent standard errors at lag 5.

3.2 Conditional density

For density forecasting we assume residuals follow the standardised skewed-t distribution of Hansen (1994):

$$\varepsilon_{t+1}/\sigma_{t+1} \sim \text{Skewed-t}(\nu, \lambda),$$

with degrees-of-freedom parameter $\nu > 2$ and skewness parameter $\lambda \in (-1, 1)$. Both are estimated jointly with the HAR parameters by maximum likelihood. The density permits closed-form expressions for value-at-risk and supports analytical Expected Shortfall computation for the standardised Student-t case.

3.3 Forecast evaluation

We evaluate point and density forecasts using three complementary criteria. The Continuous Ranked Probability Score (Gneiting & Raftery, 2007),

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{1}\{z \geq y\})^2 dz,$$

is a strictly proper scoring rule for density forecasts. The QLIKE loss of Patton (2011),

$$\text{QLIKE}_t = \log \hat{\sigma}_t^2 + RV_t / \hat{\sigma}_t^2,$$

is robust to measurement error in RV . Equal-predictive-accuracy testing uses the Diebold-Mariano (1995) statistic with Newey-West variance estimation,

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})/n}} \xrightarrow{d} N(0, 1), \quad d_t = L_t^A - L_t^B.$$

4. Data and realised-variance estimator

4.1 Sample

Our data comprise BTC-EUR mid-price candles obtained from the public Bitvavo REST API at five intraday frequencies (one-minute, five-minute, fifteen-minute, one-hour, daily) spanning 8 March 2019 to 14 May 2026. After computing daily realised measures with a 26-day warmup, the estimation panel covers 2 334 days from 3 April 2019. We reserve the period 1 October 2022 to 14 May 2026 (1 312 days) for out-of-sample evaluation, leaving 2022 in-sample days for training and refitting under a walk-forward scheme described in Section 6.

4.2 Realised-variance estimator

We compute daily realised variance using the Realised Kernel (Barndorff-Nielsen, Hansen, Lunde, & Shephard, 2008) at five-minute frequency with a Parzen kernel:

$$RK_t = \sum_{h=-H}^H k\left(\frac{h}{H}\right) \gamma_h(r),$$

where $\gamma_h(r)$ is the realised autocovariance at lag h and $k(\cdot)$ is the Parzen kernel. The five-minute frequency is selected on the basis of a signature plot (Figure 1, omitted for brevity) showing that one-minute returns exhibit a 33% upward bias from microstructure noise, reduced to approximately 10% at five-minute frequency.

4.3 Stylised facts

Six empirical features of BTC-EUR realised variance motivate the HAR-RS-DOW specification:

1. **Rough volatility** (Gatheral, Jaisson, & Rosenbaum, 2018): the Hurst parameter for log-RV increments is $\hat{H} = 0.063$, far below 0.5.
2. **Weekend effect**: weekend realised variance is 37% lower than weekday on average.
3. **Leverage**: standard sign (negative returns predict higher subsequent variance), with $\hat{\beta}_d^- > \hat{\beta}_d^+$ confirmed in estimation.
4. **Heavy tails**: Student-t maximum-likelihood degrees of freedom for raw returns is $\hat{\nu} = 3.02$.
5. **Approximate Gaussianity of log-RV**: sample skewness ≈ 0 , sample kurtosis ≈ 3.4 .
6. **Long memory**: GPH (Geweke-Porter-Hudak) fractional integration estimator yields $\hat{d} = 0.653$.

These features collectively support a model specification combining multi-horizon aggregates (HAR), asymmetric daily components (RS $\square$ /RS $\square$), and calendar effects (DOW dummies)—the configuration of HAR-RS-DOW.

5. In-sample estimation results

Table 1 reports BIC values for ten HAR-family specifications estimated on 2 312 in-sample days.

Table 1. In-sample BIC comparison (n = 2 312)

Rank	Specification	BIC	Δ BIC vs winner
1	HAR-RS-DOW	−10 462	0

Rank	Specification	BIC	Δ BIC vs winner
2	HAR-RS-Q-WE-X	−9 972	+490
3	HAR-RS-X	−9 815	+647
4	HAR-RS-WE	−9 711	+751
5	HAR-WE	−9 624	+838
6	HAR-Q	−9 481	+981
7	HAR-X	−9 332	+1 130
8	HAR-Leverage	−9 245	+1 217
9	HAR-RS	−9 138	+1 324
10	HAR (Corsi 2009)	−8 921	+1 541

A Δ BIC of 490 between HAR-RS-DOW and the next-best specification corresponds to a Bayes factor of approximately 10^{106} —overwhelming evidence by any conventional standard.

Estimated parameters for HAR-RS-DOW (Newey-West HAC standard errors in parentheses, lag = 5) are:

$$\hat{\beta}_d^+ = 0.18 (0.04), \quad \hat{\beta}_d^- = 0.34 (0.05), \quad \hat{\beta}_w = 0.31 (0.06), \quad \hat{\beta}_m = 0.21 (0.05).$$

The asymmetry $\hat{\beta}_d^- > \hat{\beta}_d^+$ confirms the Patton-Sheppard (2015) finding for an equity-like leverage effect at the variance level, here in a non-equity context. The estimated day-of-week coefficients are $\hat{\gamma}_{Mon} = +0.142$, $\hat{\gamma}_{Tue} = +0.087$, $\hat{\gamma}_{Wed} = +0.063$, $\hat{\gamma}_{Thu} = +0.054$, $\hat{\gamma}_{Fri} = +0.038$, $\hat{\gamma}_{Sat} = -0.281$. The pronounced Saturday effect reflects a structural difference in BTC liquidity around weekend periods.

6. Out-of-sample forecast evaluation

6.1 Walk-forward setup

We adopt a walk-forward evaluation with 20-day refit period: starting from the initial in-sample fit (2 022 days), the model is refit every 20 days on all data available up to that point, and produces one-step-ahead density forecasts for the following day. The resulting evaluation panel comprises 1 312 OOS observations from 1 October 2022 to 14 May 2026.

6.2 Results at $h = 1$

Table 2. Out-of-sample forecast performance, $h = 1$ ($n = 1\,312$)

Specification	CRPS ↓	QLIKE ↓	RMSE ↓	R ² _OOS ↑	DM vs HAR-RS-DOW
HAR-RS-DOW	0.498	0.595	0.889	+0.439	— (baseline)
HAR-RS-Q-WE-X	0.582	0.725	0.948	+0.246	DM = −7.34, $p < 0.001$
HAR-RS-X	0.585	0.742	0.952	+0.233	DM = −7.51, $p < 0.001$
HAR-RS-WE	0.587	0.748	0.954	+0.228	DM = −7.42, $p < 0.001$
HAR-WE	0.589	0.685	0.957	+0.225	DM = −6.89, $p < 0.001$

HAR-RS-DOW dominates all alternative specifications on every scoring rule, with Diebold-Mariano statistics rejecting equal predictive accuracy at $p < 0.001$ against all four nearest competitors. The out-of-sample

R^2 of +0.439 (computed against the in-sample mean) indicates that approximately 44% of out-of-sample log-RV variation is explained by HAR-RS-DOW's conditional forecast.

6.3 Multi-horizon results

Table 3. Multi-horizon forecast performance

Specification	CRPS h=1	CRPS h=5	R^2 h=1	R^2 h=5
HAR-RS-DOW	0.499	0.581	0.44	0.23
HAR-RS-Q-WE-X	0.582	0.638	0.25	0.16
HAR-WE	0.591	0.642	0.22	0.09

The Diebold-Mariano statistic at $h = 5$ against HAR-WE is -5.46 ($p < 10^{-7}$). The relative advantage of HAR-RS-DOW persists, with both alternatives degrading more rapidly at the weekly horizon.

7. Application: Value-at-Risk and Expected Shortfall

7.1 VaR and ES under three density specifications

Given the conditional density of next-day log realised variance, we derive the conditional density of next-day return as $r_{t+1} = \sigma_{t+1} z_{t+1}$, where z_{t+1} has unit variance and follows one of three distributions: Normal, standardised Student-t with $\nu = 3$, or Hansen (1994) skewed-t with empirically estimated $(\hat{\nu}, \hat{\lambda})$. For the standardised Student-t case, Expected Shortfall admits the closed form

$$ES_{\alpha} = -\sigma_{t+1} \cdot \frac{\nu + t_{\alpha}^2}{\nu - 1} \cdot \frac{f_{\nu}(t_{\alpha})}{\alpha} \cdot \frac{1}{\sqrt{\nu/(\nu - 2)}},$$

where t_{α} is the α -quantile and f_{ν} is the standard-t density. Maximum-likelihood estimation of the skewed-t density on standardised residuals $z_t = r_t / \hat{\sigma}_t$ yields $\hat{\nu} = 4.37$ and $\hat{\lambda} = +0.013$. The near-zero skewness of standardised residuals is notable: it indicates that the conditional volatility σ_t captures essentially all return asymmetry, leaving symmetric standardised innovations.

7.2 Backtesting Expected Shortfall

We backtest Expected Shortfall using the Acerbi and Szekely (2014) Z_1 statistic,

$$Z_1 = \frac{1}{N_{\alpha}} \sum_{t: r_t \leq \text{VaR}_{\alpha}(t)} \frac{r_t}{ES_{\alpha}(t)} + 1,$$

where the sum runs over tail-violation days. Under the null of correct model specification, $\mathbb{E}[Z_1] = 0$; values significantly above zero indicate a conservative model (realised tail losses milder than forecast), values below zero indicate underforecasting.

Table 4. Expected Shortfall backtest results (1 312 OOS days)

Density	$\alpha = 1\%$ forecast	$\alpha = 5\%$ forecast	Z1 at 1%	Z1 at 5%	Verdict
Normal	−5.93%	−4.59%	+2.106	+1.895	Conservative
Student-t ($\nu = 3$)	−8.99%	−4.97%	+1.729	+1.825	Conservative
Hansen skewed-t ($\hat{\nu} = 4.4$, $\hat{\lambda} = +0.01$)	−7.86%	−5.06%	+1.834	+1.812	Conservative
Realised	−5.84%	−3.91%	—	—	—

Across all three density specifications and both significance levels, the Z1 statistic is positive—indicating that the HAR-RS-DOW conditional density consistently overforecasts tail-loss severity. For the regulatory context this is the desirable direction: a bank deploying this model under Basel III FRTB would reserve adequate capital against tail risk.

8. Application: Black-Scholes option pricing

8.1 Setup

We construct a forward-looking valuation exercise for at-the-money straddles on BTC. On each OOS day t , three vol-input specifications are used to price a seven-day at-the-money straddle: a constant volatility (equal to the full-sample realised volatility, $\hat{\sigma}_{\text{const}}$), a backward-looking 30-day rolling realised volatility, and the HAR-RS-DOW conditional forecast $\hat{\sigma}_t$ (annualised). The Black-Scholes formula for the straddle premium with zero risk-free rate and at-the-money strike is

$$\text{Straddle}(S, \sigma, T) = 2 \cdot S \cdot [\Phi(0.5\sigma\sqrt{T}) - 0.5].$$

The realised straddle payoff at expiry is $|S_{t+7} - S_t|$. The ‘edge’ for the straddle seller is the difference between premium received and payoff paid.

8.2 Results

Table 5. Black-Scholes straddle pricing performance (n = 1 305 OOS valuations)

σ -input	Avg premium	Avg payoff	Avg edge	Hit rate	MAPE
Constant σ	€11.16	€10.30	+€0.86	66.4%	7.26
Rolling 30-day σ	€10.59	€10.30	+€0.29	63.0%	7.51
HAR-RS-DOW $\hat{\sigma}$	€10.50	€10.30	+€0.20	60.0%	7.50

All three vol-specifications yield positive average edge—a manifestation of the well-documented variance risk premium (Bakshi & Kapadia, 2003) by which options market-makers earn a premium for bearing realised-variance risk. HAR-RS-DOW pricing is closest to fair value, exhibiting an average edge of €0.20 versus €0.86 for constant volatility. The hit rate of 60% under HAR pricing is close to the theoretically efficient level for a market-maker hedging at zero edge, indicating that HAR-driven quotes are most closely aligned with realised dynamics.

9. Robustness

9.1 Cross-asset replication

To verify that the HAR-RS-DOW dominance is not artifact of the BTC-EUR sample, we replicate the in-sample horse race on ETH-EUR daily realised variance over the same period. HAR-RS-DOW retains its rank-1 position with $\Delta\text{BIC} = 387$ over the next-best specification, supporting the conclusion that the empirical advantage is structural rather than dataset-specific.

9.2 Sub-sample stability

Splitting the out-of-sample window into an ‘early’ period (2022-10 to 2024-01, $n = 455$) and ‘late’ period (2024-01 to 2026-05, $n = 857$), HAR-RS-DOW retains its Sharpe-ratio rank-1 position in CRPS in both halves: CRPS of 0.461 in the early period and 0.516 in the late period, against HAR-WE’s 0.554 and 0.601 respectively.

9.3 Density-specification robustness for Expected Shortfall

As reported in Table 4, the Acerbi-Szekely Z1 statistic remains positive across three distinct density assumptions for the standardised residual: Normal, Student-t with $\nu = 3$, and Hansen skewed-t with empirically estimated parameters. The conservative-capital conclusion is not driven by the choice of density.

10. Conclusion

This paper has established three connected empirical results for the daily realised variance of BTC-EUR. First, the HAR-RS-DOW specification—combining heterogeneous-horizon autoregression, signed semivariance, and day-of-week dummies—is statistically the dominant variance forecaster within the HAR family, both in-sample ($\Delta\text{BIC} = 490$ over the next-best alternative) and out-of-sample (CRPS = 0.498 against 0.582, with Diebold-Mariano rejection at $p < 0.001$). Second, the conditional density forecasts produced by HAR-RS-DOW yield conservatively calibrated Expected Shortfall under three plausible density specifications (Normal, Student-t, Hansen skewed-t), with Acerbi-Szekely Z1 $\square [+1.73, +2.11]$ —adequate for the Basel III Fundamental Review of the Trading Book capital framework. Third, HAR-RS-DOW’s volatility forecasts as inputs to Black-Scholes option pricing produce fair-value pricing closest to market-clearing among three vol-input specifications, with operational implications for derivative market-makers.

These results align with the broader literature view of HAR-family models: they are optimally deployed as variance forecasters underpinning risk and derivative-pricing applications, not as components of directional trading systems. The empirical advantage of HAR-RS-DOW, demonstrated here for BTC-EUR, motivates its inclusion in production risk-management infrastructure for cryptocurrency assets.

Several extensions warrant further investigation. First, the cross-asset structure of variance comovements—e.g., a multivariate HAR specification covering BTC, ETH, and macro hedges—is a natural next step. Second, the deployment of HAR- σ in actual options markets, particularly on Deribit and CME Bitcoin options venues, would test the simulation’s results against realised market-maker performance. Third, the methodology developed here may extend to other asset classes where realised-measure data is available at high frequency.

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